Problem 5

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Problem 1 *Find the error in the following "solution":*

Find $\displaystyle\int_{-2}^{2} \frac{1}{x^8 - 1}\, dx$.

let $u = x^4$

then $du = 4x^3 dx$

$$\frac{du}{4x^3} = dx$$

and $u^{1/4} = x$

$u(2) = 2^4 = 16$

$u(-2) = (-2)^4 = 16$

$$= \int_{16}^{16} \frac{1}{(u^2-1)} \cdot \frac{du}{4u^{3/4}}$$

but an integral $\int_a^a f(x)dx = 0$

so $\displaystyle\int_{-2}^{2} \frac{1}{x^8-1} dx = 0$

Explanation. The error is that the definite integral is not even defined. Specifically, the function $\dfrac{1}{x^8 - 1}$ is not defined at $x = 1$, and the limit $\displaystyle\lim_{x\to 1^+} \frac{1}{x^8 - 1} = \infty$. Other errors were also made. For example, when $-2 \leq x \leq 0$ we know that $u^{\frac{1}{4}} = -x$, not x, which makes the rest of this "solution" invalid.